



OPEN Mapping of regions with low tuberculosis notification and estimation of diagnostic gaps in Cameroon, evidence from OpenStreetMap and WorldPop data

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Since 2012, Cameroon has introduced rapid molecular diagnostic tests for tuberculosis (TB). Despite this progress, WHO estimates indicate a TB diagnostic gap of 43% at the national level in 2019. This raises questions about the strategic allocation of available rapid molecular diagnostic tools to areas with lower TB notification. In a cross-sectional study, we combined Cameroon notification data on TB (2019), rifampicin-resistant (RR)-TB (2015–2019), as well as local TB risk factors, availability, intensity of use and accessibility of the Xpert MTB-RIF test with openly available geospatial datasets from OpenStreetMap and WorldPop. A mathematical model estimated TB and RR-TB incidence rates at the regional level. We compared these estimates with the number of reported TB cases to identify diagnostic gaps. Centre, East and Far North regions had the highest estimated TB incidence rates (400, 300 and 200 cases per 100,000 inhabitants, respectively), while South and Adamawa had the highest estimated RR-TB incidence rates (14.9 and 8.9 cases per 100,000 inhabitants, respectively). We report a national diagnostic gap of 53% and 50% for TB and RR-TB, respectively. These findings highlight the need to improve the allocation of diagnostic tools that follows the local disease burden in resource-limited settings to improve health equity.

Keywords Spatial analysis, Under-diagnosis, Tuberculosis, Open-source data, Cameroon

The global decline in tuberculosis (TB) and rifampicin-resistant TB (RR-TB) incidence has lagged behind overall medical and technological advancements in high burden countries. Globally, when we compare estimations from the World Health Organization (WHO) and countries reports, we observed that 30% and 40% of TB and RR TB patients respectively, were not diagnosed¹. In 2011, WHO endorsed the use of Xpert MTB/RIF assay (Cepheid, US; Xpert) as one rapid molecular diagnostic tool (mWRD)² for TB and RR-TB, and in 2016, the TB loop-mediated isothermal amplification (TB LAMP, from Eiken, Japan) as a rapid TB-only diagnostic tool³ to be used at peripheral level of care.

Cameroon is classified as a high burden country for TB and human-immune deficiency virus (HIV)⁴, where 43% and 77% of WHO estimated TB and RR-TB patients were not diagnosed¹. The country has 28.4 million inhabitants spread over ten administrative regions. While 37.5% of the population is living under the poverty line, there is an active traditional and/or industrial mining activity ongoing in several regions⁵. Several TB risk factors facilitates the progression from TB infection to disease and their understanding is important for a better

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TB control. Additionally, late presentation and limited rapid molecular TB diagnostic facilities at the peripheral level of care in the country may lead to missed opportunities for diagnosis. Nationwide, Xpert and TB LAMP serves as near-point of care mWRD for TB diagnosis, free of charge for patients presenting at the health facility with signs and symptoms of the disease. As of 2019, 28 Xpert machines provided initial TB and RR-TB diagnosis for all presumptive patients with either paucibacillary TB (children less than 5 years old, people living with HIV) or RR-TB (patients with treatment failure and/or relapse from a previous regimen), or prisoners and foreigners from countries with a high prevalence of second-line TB drug resistance⁶. In the same year, 27 TB LAMP machines were widely available for TB-only diagnosis⁷ among adults and children > 5 years old with presumptive TB presenting at the primary level of care⁶. While all Xpert machines are linked using an embedded data connectivity system (Data-to-care from Savics, (<https://savics.org/datatocare/>)), only 26% (7/27) TB LAMP machines have a real-time activity monitoring with data connectivity. A total of 261 health facilities (public and private) served as specialised TB treatment centres, including eleven for RR-TB management.

Spatial analysis has been described by several authors as a method for integrating multiple layers of data on population demographics, health infrastructure availability, accessibility and use, and environmental factors to improve understanding of the TB epidemic and heterogeneity distribution^{8–10}. In Cameroon, it is not known whether mapping of regions with low TB notification could potentially inform national policy on the distribution of limited available rapid TB molecular diagnostic tools and thus support interventions to reduce the diagnostic gap. In this initial study, we used a multi-disciplinary approach combining openly available data on known TB risk factors for Cameroon, national TB reports (notification and pre-recorded Xpert machines activities and localization) and modelling to map regions where we estimate TB/RR-TB under-diagnosis and calculate the diagnostic gaps.

Methods

We performed a cross sectional study using openly available data for Cameroon adjusted for all the ten regions and we base our analysis on a previously validated model in the Democratic Republic of Congo¹¹. We considered all forms of TB counts reported by the local health care system aggregated per region for TB (2019) and RR-TB (2015–2019) and the pre-recorded Xpert activity nationwide. The locations of health facilities were downloaded from the OpenStreetMap project (<https://www.openstreetmap.org>) in April 2023 and we used openly accessible nationwide data on key TB risk factors (locations of mines, health infrastructure distribution, poverty) from the OpenStreetMap project, and population densities from the WorldPop project as of 2020 (<https://hub.worldpop.org/geodata/summary?id=25658>).

Reported TB and RR-TB incidence

To calculate TB and RR-TB notifications for each of the ten regions, we used a pragmatic methodology that incorporated routine notification data, the availability and accessibility of rapid molecular diagnostic tests, the intensity of their utilization, and contextual factors, as documented in previously¹². To achieve this, we first computed the time span (in quarterly intervals) during which at least one Xpert machine was operational in the region. This allowed us to track periods during which rapid molecular diagnostic capacity was available. We then determined the number of RR-TB cases reported during this operational period.

To estimate the annual number of cases, we multiplied the total number of TB and/or RR-TB cases reported within the operational time span by three, the number of months in a quarter. This approach adjusts for the operational availability of diagnostic tools and provides a more accurate estimate of TB and RR-TB notifications, accounting for diagnostic constraints and service availability within each region.

Estimated TB and RR-TB incidence rates

We used the Susceptible-Infectious-Susceptible compartmental model (which include several parameters among which population size and infection rate to describe the flow between susceptible to infectious rate) to predict TB and RR-TB incidence in each region by incorporating country available TB risk factors (proximity to health care, socioeconomic status and population density) to the model.

The following assumptions were made prior to analysis: (i) All cases were reported if living within 5 km of a hospital, (ii) half of all cases if within 5 km of a health facility different than a hospital, and (iii) one third if further away, giving three distinct distance category from healthcare; (iv) we predicted that in mining locations, the true incidences of TB and RR-TB were 1% and 0.013% respectively¹¹.

To estimate the incidence of RR-TB in mining locations, we applied a proportional relationship, where the RR-TB incidence rate is derived by multiplying the TB incidence rate in similar locations by the factor ($\#TB/\#RR-TB$) = 0.013 approximately. Key parameters in the model include the TB incidence rate, the ratio of TB to RR-TB cases, and population density, healthcare accessibility that may influence RR-TB incidence distribution.

To calculate annual estimated TB and/or RR-TB incidence, we first disaggregated notified TB and/or RR-TB cases by region, then we used a compartmental model and other data to disaggregate further to the grid level and stratifying by the three distinct distance categories from healthcare facilities. For each region, the incidence was adjusted within a 100 × 100 m grid (to align with the population data grid resolution allowing a precise spatial disaggregation), with incidence rates increasing in cells located further from healthcare access points, providing more refined incidence values across the area. Finally, to estimate total annual incidence for each region, we computed the sum of incidence $i \times$ population i , where i represents each 100 × 100 m cell across the region. For each distance category, we then sum the weighted incidence values across the cells, where each cell's weight is determined by its population density, resulting in an aggregated incidence value that considers both distance and population density. This approach allowed for a nuanced estimation of TB and/or RR-TB incidence at the regional level, sensitive to variations in healthcare accessibility and population density.

The TB diagnostic gap was calculated by comparing the true estimated incidence from the model and the actual notification for TB and/or RR-TB nationwide.

We plotted estimated and reported TB and RR-TB incidence in Cameroon in person-time year (per 100 000 inhabitants) using regional geographic points (all the ten regions). We added a 95% confidence interval on our estimated TB and RR-TB incidence to compare results from the model versus reported TB and RR-TB incident cases per region.

Ethical approval

Our study did not require ethical approval as we used openly available data only. For the same reason, we did not obtain informed consent from patients.

Results

We identified 716 health facilities (including 217 clinics and 206 hospitals) spread over the country as of April 2023 (additional Fig. 1).

Estimated and reported TB incidence in Cameroon

The largest estimated low-TB notifications were observed in the East, Centre and Far North regions (Fig. 1).

In the left panel, areas with low population density appear as disease free with white colour meaning zero incidence (as expected by the underlying model).

Estimated and reported RR TB incidence in Cameroon

RR TB incidence rates varied considerably across regions of Cameroon, with the South and Adamawa regions displaying the highest number of new RR TB patients for both estimated and reported (Fig. 2).

The model estimated a yearly average of 52,273 TB patients over the working year of Xpert machines in the country, while only 24,582 patients were actually reported, indicating a significant diagnostic gap of around 53%. Similarly, for RR-TB, the model estimated a yearly average of 644 patients over the working year of Xpert, while only 320 patients were reported, indicating a diagnostic gap of around 50% (Fig. 3).

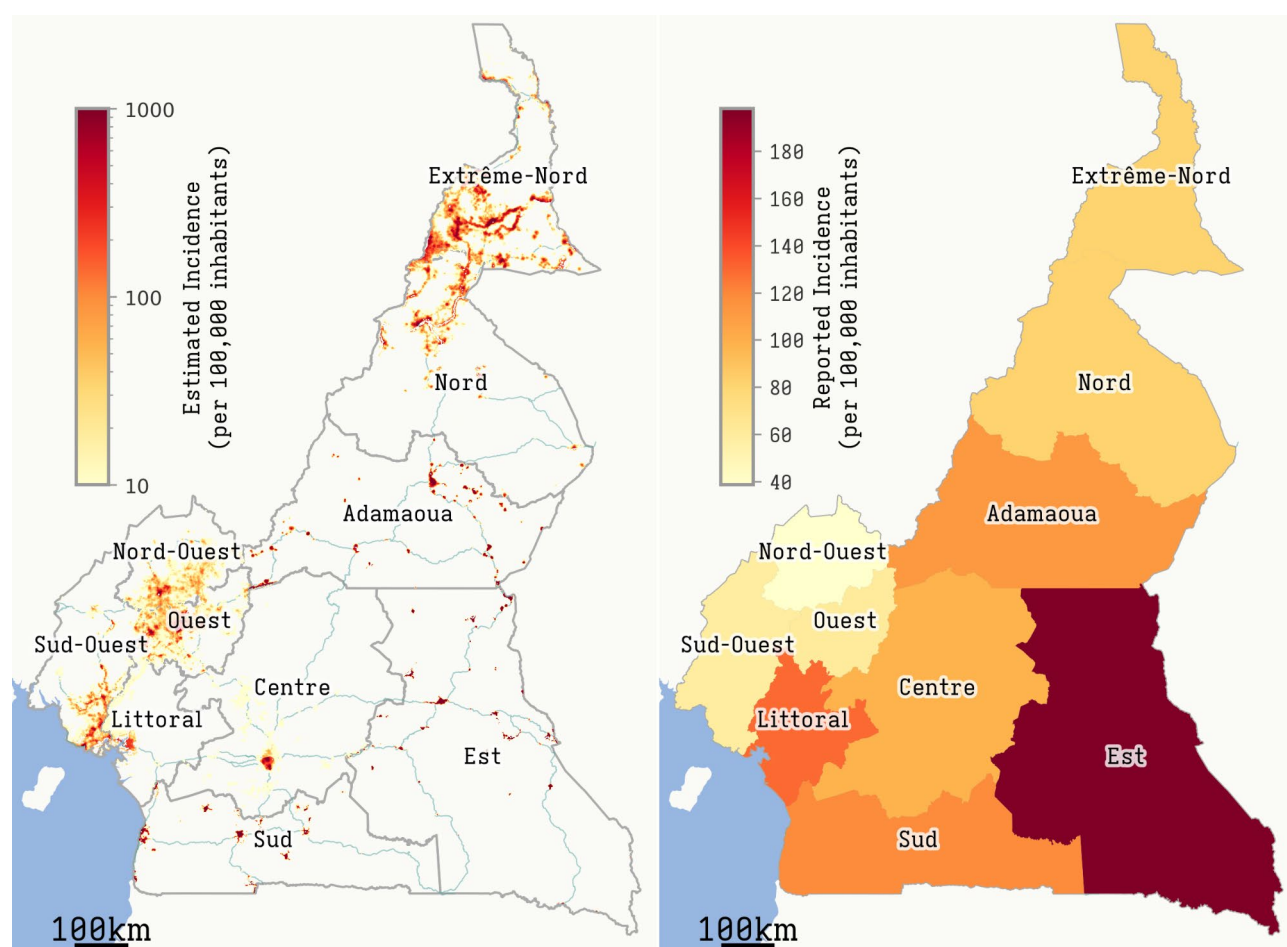


Fig. 1. Estimated TB incidence (left) by our approach and reported by the local health care system (right) in Cameroon. In the left panel, areas with low population density appear as disease free with white colour meaning zero incidence (as expected by the underlying model).

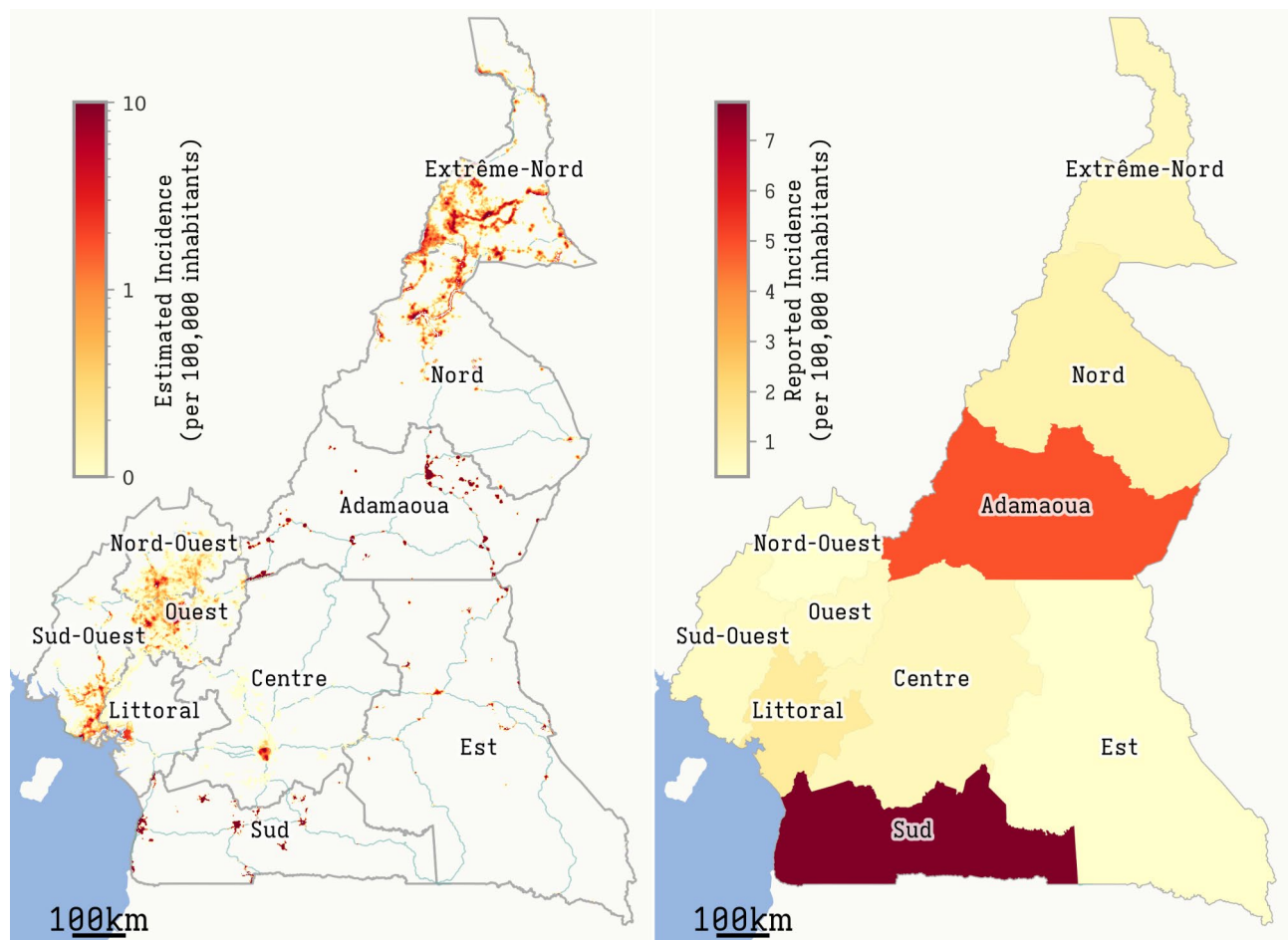


Fig. 2. Estimated RR TB incidence (left) by our approach and reported by the local health care system (right) in Cameroon. In the left panel, areas with low population density appear as disease free with white colour meaning zero incidence (as expected by the underlying model).

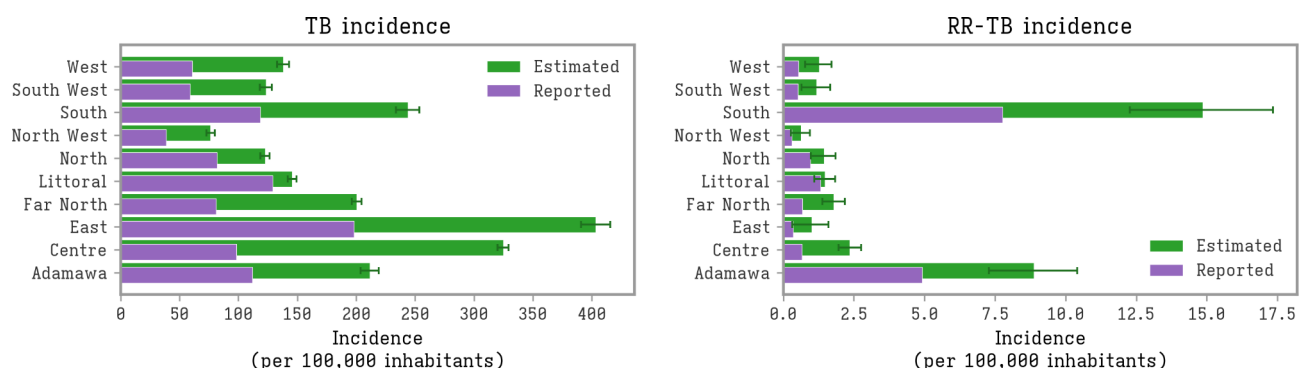


Fig. 3. Estimated TB (left) and RR TB (right) incidence versus reported per region with confidence intervals.

Average TB and RR TB estimated incidence rates in Cameroon

Regions of East, Centre and Far North displayed higher TB average estimated incidence respectively at 400, 300 and 200 per 100,000 inhabitants. While the South and Adamawa regions displays higher RR-TB estimated incidence at 14.9 and 8.9 per 100,000 inhabitants, respectively (Fig. 4). Notably, some regions estimated to have high average TB incidence rates (Centre, East) were shown to have low average RR-TB incidence rates.

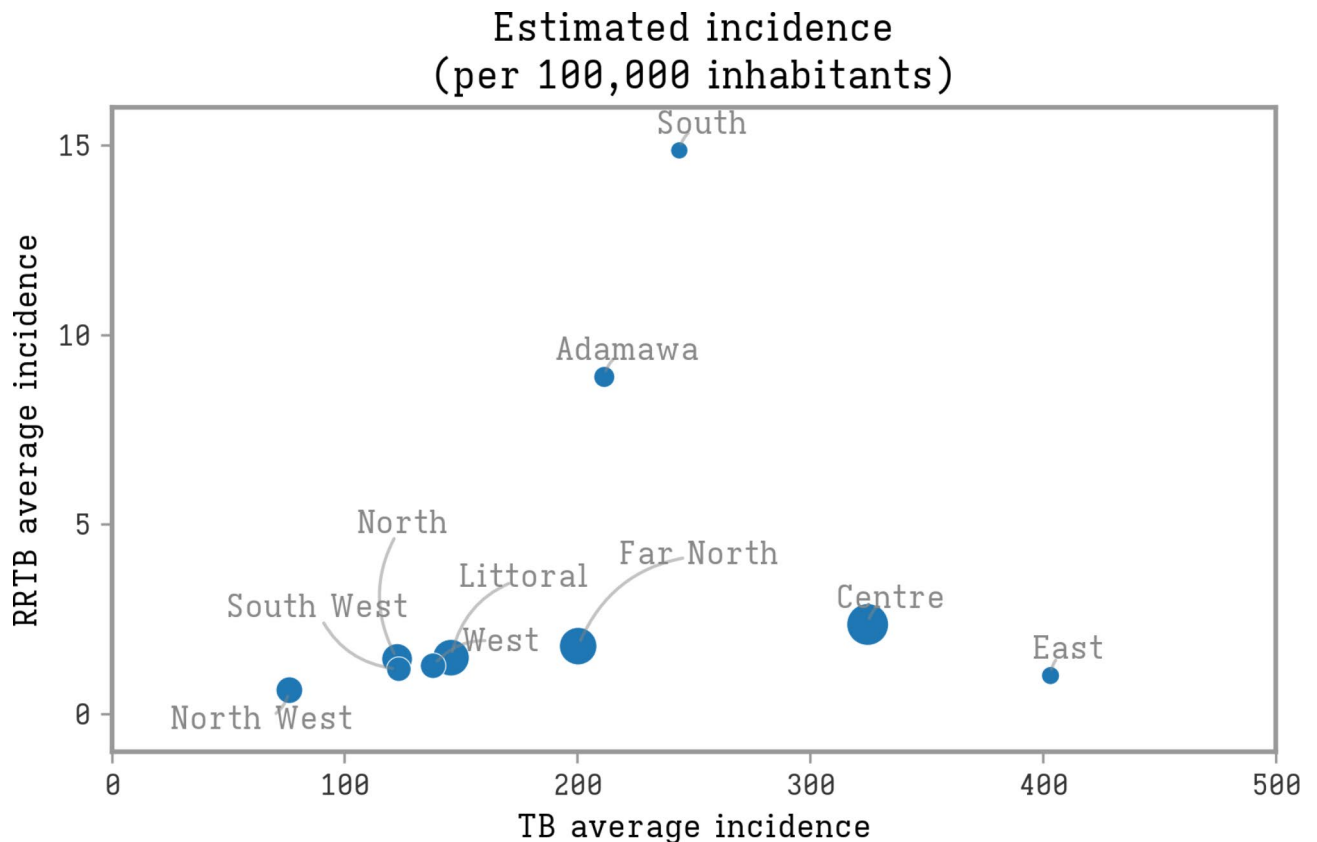


Fig. 4. Average TB and RR-TB notification and estimated incidence rates per region in Cameroon.

Discussion

This initial analysis from Cameroon integrate in a comprehensive way, openly available data on TB risk factors, TB reports (notification and pre-recorded Xpert MTB/RIF activity in all the regions) and mathematical modelling to map areas where we estimate to find more TB and RR TB patients than what is reported by the local health system in order to reduce the diagnostic gap.

We estimated that approximately 53% of the total TB patients were missed by the local health system in 2019, close to 56% reported by WHO for the same year in Cameroon; while for RR-TB we estimated that approximately 50% of patients were not diagnosed by routine program interventions. This is less than the 77% gap reported by WHO TB report for the country¹³. The overall observed regional discrepancies between estimated and reported TB and RR-TB incidence rates in some regions can be attributed to several factors. First, the use of notification data in our analysis which depends on patients developing the signs and symptoms of the disease then presenting at a specialized TB health facility to have the correct Xpert diagnosis and reporting. Along this care cascade, patients may experience delays either in seeking care to the TB specialized health facility or due to logistical and operational challenges inherent to Xpert (module failing for calibration, shortage of cartridges, unstable internet)^{14,15}. These factors could lead to over-or -under estimation of the true incidence rates therefore impacting the accuracy of the identified regions with low TB notifications. Secondly, we assumed that all TB patients were notified if they were living 5 km closer to a hospital, half of all cases if within 5 km of a health facility different than a hospital, and one third if further away. These assumptions may not reflect the Cameroonian context, where access to TB services at the primary level of care ranges from 3 to 16%, and it is where 87% of the population do seeks primary care¹⁶. Moreover, the model does not account for cross-regional patient movement for diagnosis and treatment, which might skew the estimated TB incidence in certain areas. Finally we speculate that high-risk populations, such as miners and people living with HIV, may face stigma and discrimination which further exacerbate underreporting in some regions.

Our study had several limitations that should be acknowledged. One significant limitation is the exclusion of TB-LAMP activity data from the analysis which may provide a more comprehensive picture of TB incidence if integrated into the model. Second, the analysis of estimated TB-only incidence per region was based on TB notification data in 2019, by including reported TB data over several years in the model, policy makers will have a better picture of TB incidence in Cameroon. Third, relying solely on reported cases may have introduced bias as it does not capture the full spectrum of the disease, including unreported TB cases in the country. Nevertheless, our study illustrates that geospatial techniques could provide a more accurate analysis of areas with lower TB notification in Cameroon, enabling more precise allocation of available rapid molecular diagnostic tools for a better TB control. To improve the robustness of this analysis, we are planning to include data on more refine TB

risk factors (HIV, age and sex distribution per region), prospectively collected data of the TB LAMP activity, and TB notification over time, this refined analysis could provide a more accurate understanding of TB incidence in Cameroon, taking into account unreported TB cases.

As global priorities have shifted from controlling to ending tuberculosis by 2035, this advanced analytical approach offers a promising avenue for catching up with modern technological tools and addressing the diagnostic gaps that have hindered effective TB control in high-burden countries.

In conclusion, the substantial 53% and 50% of TB and RR-TB diagnostic gaps reported in this analysis, underscore the urgent need for region-specific interventions to enhance TB diagnosis nationwide. Tailored diagnostic strategies and optimized use of available diagnostic tools are essential to bridge these diagnostic gaps for both TB and RR TB to improve overall TB control efforts nationwide.

Data availability

The datasets generated and/or analysed during the current study were downloaded from the public database of countries (<https://download.geofabrik.de/>) and we used overpass for the data of interest (<https://github.com/mvexel/overpass-api-python-wrapper>).

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Author contributions

CGJ: Conception, design of the work, acquisition, interpretation of data, drafted the work and approved the final version of the work, agreed to be accountable for author's own contributions. TNC: Acquisition of the data, Approved the final version of the work, agreed to be accountable for author's own contributions. BT: Analysis of the data, Approved the final version of the work, agreed to be accountable for author's own contributions. CG: Substantively revised the work, approved the final version of the work, agreed to be accountable for author's own contributions. PMN: Interpretation of data, Approved the final version of the work, agreed to be accountable for author's own contributions. EA: Conception, design of the work, Interpretation of data, Creation of the new model used in the work, Substantively revised the work, Approved the final version of the work, agreed to be accountable for author's own contributions. MF: Analysis of the data, Interpretation of data, Creation of the new model used in the work, Substantively revised the work, Approved the final version of the work, agreed to be accountable for author's own contributions.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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